

Lathmi Wanigasekara

Software Engineering Undergraduate

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About Me

Second-year Software Engineering undergraduate with hands-on experience developing scalable backend systems, RESTful APIs, and database-driven web applications using FastAPI, Node.js, PostgreSQL, and MongoDB. Experienced in building real-time IoT-integrated solutions with JWT authentication, analytics systems, and Agile-based development workflows. Strong foundation in software engineering principles, clean architecture, and collaborative development practices.

Technical Skills

- Programming Languages: JavaScript, Python, Java, PHP
- Frontend: React.js, Next.js, HTML5, CSS3, EJS templates, Flutter
- Backend & API Development: FastAPI, Node.js, Express.js, RESTful APIs
- DevOps & Cloud: Docker (learning), AWS (fundamentals), Linux CLI, Bash scripting
- Databases: PostgreSQL, MySQL, MongoDB, SQLAlchemy (ORM), Firebase, Supabase
- Authentication & Security: JWT Authentication, OAuth 2.0
- Testing & Tools: JUnit, Postman, Swagger
- Version Control: Git, GitHub
- Development Methodologies: Agile, Scrum

Projects

AquaSense – Smart Water Management, Leak Detection and Control System (2025-2026)

- Built backend REST API using FastAPI and Python
- Designed PostgreSQL database and ORM models using SQLAlchemy
- Implemented JWT authentication and user management system
- Integrated real-time communication between IoT devices and server
- Implemented water usage analytics, leak detection, and daily summaries
- Managed software development tasks using ClickUp.
- Coordinated backend, frontend, database, and IoT development tasks
- Followed Agile-based project workflow.
- Monitored project progress and task completion

To-Do List Web Application (2026)

- Built using Node.js, Express, MongoDB with MVC architecture
- Developed dynamic UI using EJS templates
- Implemented RESTful routing and full CRUD operations
- Managed data flow between frontend and backend with efficient routing and rendering

Acyclicity Checker - Directed Graph Algorithm Implementation (2026)

- Developed a Java-based acyclicity checker for directed graphs using a sink-elimination algorithm with $O(V+E)$ time complexity.
- Implemented an iterative DFS cycle detection algorithm with three-colour marking to identify and reconstruct cycle paths while avoiding stack overflow on large graphs.
- Designed a custom adjacency-list graph structure using dual HashMaps for efficient edge management and graph traversal.
- Diagnosed and resolved a cycle path reconstruction bug, improving algorithm accuracy and reliability.
- Validated algorithm performance through complexity testing across multiple graph sizes.

E-commerce Web Application (2025)

- Developed an e-commerce system using PHP with session-based cart management
- Implemented product handling and dynamic page updates
- Focused on backend logic and efficient user session handling

Life Below Water Website Design and Development (2024)

- Built a responsive web app promoting sustainability awareness
- Collaborated in an Agile team, contributing to both frontend design and development
- Ensured cross-device compatibility and user-friendly UI/UX

Java Theatre Management System (2025)

- Developed a modular theatre management system using object-oriented programming principles
- Implemented show scheduling, user registration, membership management, and rewards functionality
- Applied clean system design concepts for maintainable and scalable application structure

Education

BSc (Hons) Software Engineering (2024 – 2028)

University of Westminster (Informatics Institute of Technology, Sri Lanka)

Machine Learning – Professional Certificate Programme (32 Hours)

Informatics Institute of Technology – Merit Pass

Foundation Certificate in Higher Education (2023-2024)

Informatics Institute of Technology

G.C.E Ordinary Level (2019)

Eight 'A' passes and one 'B' pass.

Leadership/Activities

- Assistant secretary of the Interact Club of Viharamahadevi Balika Vidyalaya (2021-2022)
- Junior Prefect of Viharamahadevi Balika Vidyalaya (2016)
- Member - Western Band, Choir, Recorder Group, and Mixed Instruments Group